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DF^2-VB: Dual-level Fuzzy Fusion with View-specific Boosting for Multi-view Multi-label Classification

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Abstract

Multi-view multi-label classification (MVMLC) aims to utilize both consensus and complementarity information to predict potentially relevant labels for samples. Existing MVMLC approaches typically focus on either feature-level fusion, which integrates complementary features for more expressive representations, or decision-level fusion, which aggregates view-specific predictions to exploit label supervision more effectively. In fact, relying solely on feature-level fusion often underutilizes label information and limits discriminability of learned representations, whereas pure decision-level fusion pays insufficient attention to view representation expressiveness and thus constrains classification performance. To address these limitations, we propose DF^2-VB, a dual-level fusion framework that jointly exploits complementary strengths to mitigate their respective weaknesses by integrating feature- and decision-level fusion. At the feature fusion level, a Fuzzy Dynamic Fusion (FDF) module maps consensus features into a more compatible fuzzy feature space, where essential features are identified and redundant features are suppressed to further fuse an expressive consensus representation and boost view-specific predictions for decision-level fusion. At the decision fusion level, a View-specific Boosting (VB) strategy adaptively measures the importance of samples and view-specific predictions to strengthen the utilization of supervision for facilitating the discriminability in feature-level fusion. Complementarily, FDF and VB jointly reinforce the model expressiveness and discriminability for reliable predictions. Extensive experiments on multiple public datasets verify the superiority of our strategy over advanced MVMLC models.

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1. Introduction

Multi-view learning aims at unifying information from multiple heterogeneous types of data, where each type of data is regarded as a view [12, 25, 26, 30]. Multi-view multi-label classification (MVMLC) is one complex task in multi-view learning, which focuses on combining information from multi-view data and recognizing the multiple possible labels associated with samples [4, 13, 15, 20, 38]. For example, in emotion analysis [3], various kinds of view data are utilized to predict possible emotion labels jointly, such as scared, sad, and angry. In view of such meaningful applications in the real world, MVMLC has received increasing attention and become a burgeoning research topic [10, 19, 31].

The core of MVMLC lies in mining essential information from heterogeneous views and learning features that accurately capture label semantics [24, 39]. To this end, two mainstream MVMLC paradigms are proposed [24], namely feature-level fusion and decision-level fusion. Feature-level fusion approaches aim to integrate multiple view features into a unified feature space [5, 34]. To alleviate conflicts among heterogeneous view features, many methods introduce view-level weighting mechanisms during the feature fusion process [8, 22, 28]. Although these methods enhance feature expressiveness in practice, *they often underutilize supervision information and limit the discriminability of the learned features*. In contrast, decision-level fusion methods directly interact with supervision to enhance reliability by combining predictions from multiple view-specific classifiers [27]. Despite showing improved predictive ability, *they typically pay insufficient attention to improving feature expressiveness for further supporting predictions*.

Overall, either paradigm exposes the inherent limitations: feature-level fusion underutilizes discriminative supervision guidance, while decision-level fusion depends on weakly expressive features. Fortunately, the strengths of the two paradigms neutralize the limitations of each other: feature-level fusion can generate more expressive repre-

sentations by alleviating inter-view inconsistency conflicts, whereas decision-level fusion can enhance the discriminability of representations through effectively leveraging supervision. Motivated by this, we propose a novel model, DF^2-VB, which mitigates the weaknesses while forming expressive and discriminative representations by unifying both fusion levels. First, DF^2-VB employs pairs of consensus and complementarity encoders for each view to extract consensus and complementarity representations, respectively. Then, in the feature-level fusion phase, the Fuzzy Dynamic Fusion (FDF) module is proposed to map the consensus representations from different views into a more compatible fuzzy representation space. Unlike traditional methods that adopt view-level weighting, FDF employs membership functions to enable identification and preservation of informative features while suppressing redundant ones to yield an expressive fused consensus representation, which is then injected into complementarity representations to bolster view-specific predictions for decision-level fusion. Finally, in the decision-level fusion, we decompose the multi-label classification task into multiple simplified binary subproblems and design a View-specific Boosting (VB) strategy that evaluates the importance of samples and predictions, allowing collaborative decision-making across data with different learning difficulties and further exploiting supervision to reinforce the representation discriminability during the feature-level fusion phase. By tying FDF to VB, DF^2-VB effectively makes up for the shortcomings of both fusion, achieving more reliable predictions. Our contributions are summarized as follows:

- We propose DF^2-VB, a novel unified dual-level fusion model, to jointly optimize feature-level and decision-level fusion to neutralize their shortcomings for realizing reliable predictions.
- We design the Fuzzy Dynamic Fusion to learn adaptive feature-level weights, and develop the View-specific Boosting to dynamically measure the importance of samples and classifiers, which effectively learn expressive representations and exploit supervision information.
- Experiments on six public benchmarks demonstrate that DF^2-VB achieves consistent state-of-the-art performance and improves the expressiveness and discriminability of produced representations.

2. Related work

2.1. Multi-view Multi-label Learning

Multi-view multi-label learning (MVML) extends multi-view learning by simultaneously handling multiple heterogeneous views and multiple correlated labels [16, 38, 39]. Existing MVML approaches can be grouped into two major types: feature-level fusion-based [7, 29] and decision-level fusion-based methods [3, 6, 39]. Feature-level fusion methods focus on learning unified and informative representations from multiple views. SIMM [29] jointly extracts shared and view-specific information, and integrates them into a fused representation for final prediction. E2FS [7] enhances feature selection by jointly performing inter-view fusion and feature enhancement to achieve effective MVMLC. Although these methods improve representational expressiveness, they generally underutilize supervision information, which limits the discriminability. In contrast, decision-level fusion approaches combine view-specific predictions to achieve reliable classification. ML-BVAE [3] adopts a product-of-experts mechanism to integrate decisions from multiple classifiers within an inference network, enabling simultaneous prediction of multiple emotional states. I2VSLC [6] explores inter-view label correlations by establishing hierarchical relationships among view-specific label matrices. Despite their success, existing MVMLC methods often focus solely on either feature-level or decision-level fusion. This separation prevents them from leveraging the complementary advantages of both paradigms, and motivates us to develop a unified framework that jointly exploits the benefits of feature-level and decision-level fusion for more effective multi-view multi-label classification.

2.2. Fuzzy Set Theory

Fuzzy set theory extends classical (crisp) set theory by introducing the concept of membership degrees, enabling a more fine-grained characterization of whether an element belongs to a set [32]. A fuzzy set $\mathcal{A}$ defined on a universe $\mathbb{U}$ is represented by a membership function $f_{\mathcal{A}}(a)$, which assigns each element $a \in \mathbb{U}$ to a real value in the interval $[0, 1]$. Here, $\mathbb{U}$ denotes the collection of all elements under consideration for a specific problem, and $f_{\mathcal{A}}(a)$ quantifies the degree to which a belongs to the fuzzy set $\mathcal{A}$ [37]. Fuzzy set theory provides a flexible framework for modeling uncertainty. However, the design of appropriate membership functions is often domain-dependent and requires prior knowledge. Formally, the definitions are given by

Definition: A function $f_{\mathcal{A}} \colon \mathbb{U} \to \mathbb{L} \subseteq \mathbb{R}$ is called a membership function.

Definition: A tuple $\mathcal{A} = (\mathbb{U},\ f_{\mathcal{A}})$ including a universe $\mathbb{U}$ and a membership function $f_{\mathcal{A}}$ is called a fuzzy set.

Following above definitions, when $\mathbb{L} = [0,\ 1]$, $\mathcal{A}$ is a fuzzy set. When the membership function takes on only two values 0 and 1, namely $\mathbb{L} = \{0,\ 1\}$ according to whether a belongs to $\mathcal{A}$. In this case, $\mathcal{A}$ degenerates into a crisp set.

3. Proposed Method

Notations. In this paper, the feature space is defined as $\mathcal{X} = \mathbb{R}^{d_1} \times \mathbb{R}^{d_2} \times \cdots \times \mathbb{R}^{d_V}$, where V is the number of views, and the v-th view feature has d_v dimensions. For a given dataset $\mathcal{D} = \{(X_i, \boldsymbol{y}_i)\}_{i=1}^{N}$ with N samples, we

denote each sample $X_i = [\boldsymbol{x}_i^1, \boldsymbol{x}_i^2, \cdots, \boldsymbol{x}_i^V]$ is composed of V view features, $\boldsymbol{y}_i = [\boldsymbol{y}_{i,1}, \boldsymbol{y}_{i,2}, \cdots, \boldsymbol{y}_{i,L}]$ as the true label for X_i, where L is the number of classes. $\boldsymbol{y}_{i,j} = 1\,(1 \leq j \leq L)$ indicates that sample X_i is annotated with label j, $\boldsymbol{y}_{i,j} = 0$ otherwise. The purpose of our proposed DF^2-VB is to integrate diverse features from different views to construct a powerful multi-label classifier that can assign predictive labels for unseen samples.

3.1. Overview

General MVMLC methods jointly exploit consensus and complementarity representations to train multi-label classifiers. However, the direct combination of both is challenging: consensus representations maintaining similarity across views yield weakly discriminative predictions, whereas complementarity representations contain heterogeneous information that often disrupts feature fusion and restricts representation expressiveness. Considering this, DF^2-VB explores an effective strategy including a feature extraction and decoupling process, a feature-level fusion process, and a decision-level fusion process. In the feature-level fusion process, each view feature is decomposed into a consensus representation and a view-specific complementarity representation with the decoupling loss $\mathcal{L}_{de}$. In the feature-level fusion process, a Fuzzy Dynamic Fusion (FDF) module is proposed to map the consensus representations into a more compatible fuzzy representation space, where the feature importances are measured to identify essential features and suppress redundant ones for fusing them into a unified comprehensive consensus representation $\hat{\boldsymbol{C}}$. In the decision-level fusion phase, the fused comprehensive consensus representation is injected into each complementarity representation to obtain final view-specific representations $\{\boldsymbol{\mathcal{H}}^v\}_{v=1}^V$, which serve as inputs to generate view-specific predictions $\{\tilde{\boldsymbol{O}}^v\}_{v=1}^V$. To further refine predictions, each view-specific classifier is decomposed into multiple atom classifiers. A View-specific Boosting (VB) strategy is then introduced to adaptively weigh both atom classifiers and training samples, thereby enhancing collaborative predictions across views and enhancing the discriminability of representations. The framework is shown in Fig. 1.

3.2. Consensus and Complementarity Extraction

It has been reported that the combination of consensus and complementarity is beneficial for improving MVMLC model performance [11, 29]. Following previous works, we extract consensus and complementarity representations from various view data with different dimensions by introducing individual encoders to extract corresponding information from each view data. The process is defined as

$$\boldsymbol{C}^v = \mathcal{E}_c^v(\boldsymbol{X}^v), \quad \boldsymbol{P}^v = \mathcal{E}_p^v(\boldsymbol{X}^v), \tag{1}$$

where $\boldsymbol{X}^v$ is the feature matrix of v-th view. $\mathcal{E}_c^v$ and $\mathcal{E}_p^v$ are two encoders for extracting essential consensus representation $\boldsymbol{C}^v$ and complementarity representation $\boldsymbol{P}^v$ from the v-th view feature $\boldsymbol{X}^v$, respectively.

Since the view-specific complementarity representations are likely to cause view conflicts due to the inherent heterogeneity, while consensus representations are supposed to maintain similarity across all the view features, it is more appropriate to fuse either separately instead of fusing both directly. To this end, we decouple complementarity and consensus representations to break their interdependence for promoting the subsequent dual fusion process. The decoupling loss is defined as

$$\mathcal{L}_{de} = -\frac{1}{V}\sum_{v=1}^{V}\log\frac{\ell_1^v}{\ell_1^v + \ell_2^v + \ell_3^v}, \tag{2}$$

where

$$\begin{cases} \ell_1^v = \sum_u e^{\vartheta(\bar{\boldsymbol{c}}^v, \bar{\boldsymbol{c}}^u)/\tau_{ind}} \\ \ell_2^v = \sum_u e^{\vartheta(\bar{\boldsymbol{c}}^v, \bar{\boldsymbol{p}}^u)/\tau_{ind}} \\ \ell_3^v = \sum_u e^{\vartheta(\bar{\boldsymbol{p}}^v, \bar{\boldsymbol{p}}^u)/\tau_{ind}} \\ \bar{\boldsymbol{c}}^v = \frac{1}{N}\sum_{i=1}^N \boldsymbol{c}_i^v \\ \bar{\boldsymbol{p}}^v = \frac{1}{N}\sum_{i=1}^N \boldsymbol{p}_i^v \end{cases}. \tag{3}$$

Here, $\vartheta(\cdot,\cdot)$ is defined as the cosine similarity, τ_{ind} is a temperature coefficient, $\boldsymbol{c}_i^v$ is the i-th row of the consensus representation matrix $\boldsymbol{C}^v$, and $\boldsymbol{p}_i^v$ is the i-th row of the complementarity representation matrix $\boldsymbol{P}^v$. Different from common ways to realize decoupling with instance-level loss [17, 18], we implement mean pooling to obtain the summary consensus representation $\bar{\boldsymbol{c}}^v$ and summary complementarity representation $\bar{\boldsymbol{p}}^v$ for the decoupling process, which reduces the complexity $\mathcal{O}(V^2N^2)$ to $\mathcal{O}(V^2)$.

Theorem 3.1 *For $v, u \in \{1, \cdots, V\}$, $\mathcal{L}_{de}$ can increase the lower bound of $I(\boldsymbol{C}^v; \boldsymbol{C}^u)$ while decreasing the lower bound of $I(\boldsymbol{C}^v; \boldsymbol{P}^u)$ and $I(\boldsymbol{P}^v; \boldsymbol{P}^u)$.*

This theorem means that Eq. (2) enhances the statistical dependence of consensus representations, while suppressing both the statistical dependence of consensus and complementarity representations, and the statistical dependence of complementarity representations.

3.3. Fuzzy Dynamic Fusion

Most feature-level fusion-based methods realize feature fusion by assigning specific weights for different views [23]. Though alleviating the conflicts among view features to some extent, these methods ignore to handle the feature-element-level significance and redundancy for further fusion facilitation. Hence, we propose the fuzzy dynamic fusion (FDF) strategy to deal with these issues by adaptively measuring the significance of feature elements.

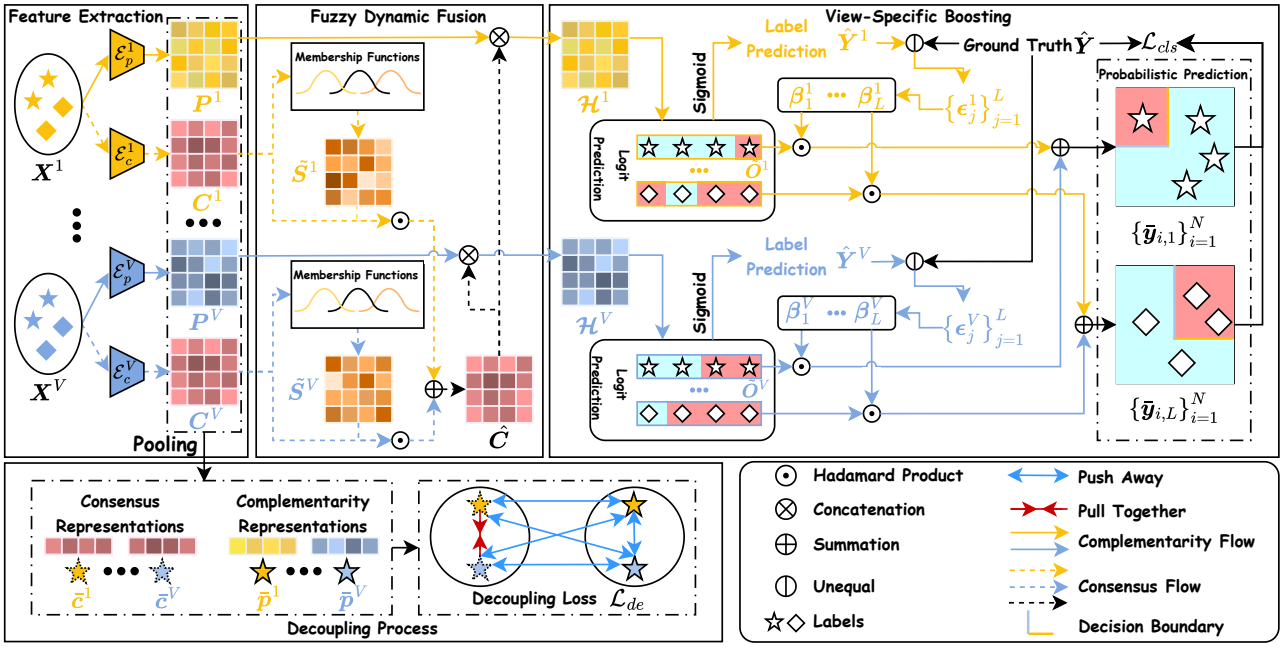


Figure 1. The framework of DF^2-VB, which consists of three processes: (1) Feature extraction and decoupling process, which decompose each view data into independent consensus and complementarity representations; (2) Feature-level fusion process, where a Fuzzy Dynamic Fusion module measures feature importances for fusing a comprehensive consensus representation; (3) Decision-level fusion process, where a view-specific Boosting strategy measures the importance of each atom classifier to build reliable decision boundaries.

FDF applies fuzzy logic [32, 33] to transform all the consensus representations into a fuzzy representation space. Since the fuzzy representation space can be considered as a more generalized feature space when compared with the normal feature space [35], it is more compatible to combine the features from heterogeneous views. To realize the transformation, we introduce K Gaussian membership functions [9, 36] for enriching the semantics information of the consensus representations as

$$\boldsymbol{s}_{i,j,k}^{v} = \exp\left\{-\frac{\left(\boldsymbol{c}_{i,j}^{v} - \boldsymbol{m}_{j,k}\right)^2}{2\boldsymbol{\delta}_{j,k}^2}\right\}, k = 1, 2, \cdots, K, \quad (4)$$

where $\boldsymbol{c}_{i,j}^{v}$ is the j-th feature element of $\boldsymbol{c}_i^v$. $\boldsymbol{m}_{j,k}$ and $\boldsymbol{\delta}_{j,k}$ are learnable mean and standard deviation of the k-th Gaussian membership function. $\boldsymbol{s}_{i,j,k}^{v}$ is the membership degree measuring how much $\boldsymbol{c}_i^v$ matches the k-th fuzzy representation space, which is considered as feature importance. Obviously, the feature importance is generated from a more fine-grained element-wise perspective. With the learnable membership functions, the feature importance is adaptively adjusted to preserve essential information while suppressing meaningless ones, which better evaluates the significance and redundancy of feature elements than methods assigning view weights.

Among K membership degrees, we pick out the final one with the max-pooling operator as

$$\tilde{\boldsymbol{s}}_{i,j}^{v} = \max_{k}\{\boldsymbol{s}_{i,j,k}^{v}\}, k = 1, 2, \cdots, K. \quad (5)$$

This degree is in line with the most informative consensus feature element, and it further improves representation expressiveness and facilitates the fusion process. When obtaining the weights for consensus representations, we then fuse them together for a comprehensive consensus representation as

$$\hat{\boldsymbol{C}} = Norm\left(\sum_{v=1}^{V} \tilde{\boldsymbol{S}}^{v} \odot \boldsymbol{C}^{v}\right), \quad (6)$$

where $\tilde{\boldsymbol{S}}^{v}$ is the membership degree matrix whose elements are obtained via Eq. (5), $\odot$ is the Hadamard product, and $Norm(\cdot)$ is the normalization operator.

Theorem 3.2 *The feature-level weighting strategy FDF can yield smaller Rademacher Complexity [1] than general view-level weighting approaches.*

Smaller Rademacher Complexity corresponds to better generalization, and it clarifies the merit of FDF when compared with mainstream view-level weighting approaches.

3.4. View-specific Boosting

View-specific complementarity representations are critical contents for reliable predictions, but they often contain heterogeneity to prevent smooth feature fusion. In view of this,

we combine the complementarity information from all the views in the decision-level fusion phase. However, only view-specific complementarity information is insufficient to guide accurate predictions since the indispensable consensus information has been stripped away from the view features. To complete the beneficial information for supporting predictions of base classifiers, we merge the view-specific complementarity representations and the comprehensive consensus representation as the final representation for each view by

$$\boldsymbol{\mathcal{H}}^v = \hat{\boldsymbol{C}} \| \boldsymbol{P}^v, \tag{7}$$

where $\|$ is the concatenation operator, $\boldsymbol{P}^v$ is the complementarity representation of the v-th view, and $\boldsymbol{\mathcal{H}}^v$ is the final representation for the v-th view.

Since individual view representations embody specific knowledge relevant to labels, it is critical to explore corresponding activation relationships between view representations and labels. To reveal the detailed activation relationships, we factorize the multi-label classification problem into multiple fine-grained binary classification subproblems. Accordingly, the prediction of each view-specific base multi-label classifier is broken up into the predictions of atom classifiers as

$$\begin{aligned} \tilde{\boldsymbol{o}}_i^v = g^v(\boldsymbol{\mathcal{H}}_i^v) = [\tilde{\boldsymbol{o}}_{i,1}^v, \tilde{\boldsymbol{o}}_{i,2}^v, \cdots, \tilde{\boldsymbol{o}}_{i,L}^v], \\ \tilde{\boldsymbol{y}}_{i,j}^v = \sigma(\tilde{\boldsymbol{o}}_{i,j}^v), \end{aligned} \tag{8}$$

where $g^v(\cdot)$ is the view-specific base classifier for the v-th view, $\boldsymbol{\mathcal{H}}_i^v$ is the final representation of the i-th sample for the v-th view, and $\sigma(\cdot)$ is the Sigmoid function. $\tilde{\boldsymbol{o}}_{i,j}^v$ is the logit of an atom classifier of the v-th view for predicting the j-th label. $\tilde{\boldsymbol{y}}_{i,j}^v \in (0,1)$ is the confidence prediction of the atom classifier.

In the decision-level fusion phase, we introduce a view-specific boosting (VB) strategy to form strong classifiers by sufficiently incorporating the advantages of each atom classifier. The key point of VB is to determine the importances of samples and atom classifiers, which encourages to calculate the error rate for each atom classifier in a mini-batch B as follows:

$$\begin{aligned} \boldsymbol{\epsilon}_{j,t}^v = \frac{\sum_i \boldsymbol{\omega}_{i,t} \cdot \mathbb{I}(\hat{\boldsymbol{y}}_{i,j}^v \neq \boldsymbol{y}_{i,j})}{\sum_i \boldsymbol{\omega}_{i,t}}, \\ \hat{\boldsymbol{y}}_{i,j}^v = \mathbb{I}(\tilde{\boldsymbol{y}}_{i,j}^v > \zeta), \end{aligned} \tag{9}$$

where $\boldsymbol{\omega}_{i,t}$ is the weight of i-th sample in the t-th boosting iteration, and $\boldsymbol{\omega}_{i,0}$ is initialized as $\frac{1}{|B|}$. $\mathbb{I}(\cdot)$ is the indicator function for counting the wrong predictions in the mini-batch B, and ζ is a threshold. $\hat{\boldsymbol{y}}_{i,j}^v \in \{0,1\}$ is the label prediction of the j-th atom classifier for the v-th view feature. $\boldsymbol{\epsilon}_{j,t}^v$ is the error rate for the atom classifier that predicts the j-th label with $\boldsymbol{\mathcal{H}}^v$.

When obtaining the error ratios, the strengths of atom classifiers to recognize the activation relationships between representations and labels are clear, *i.e.*, the correspondences between labels and view representations are identified. The importances of atom classifiers are controlled by the error ratios according to the following rule:

$$\begin{aligned} \boldsymbol{\beta}_{j,t+1}^v = \begin{cases} \frac{1}{2}\log\frac{1-\boldsymbol{\epsilon}_{j,t}^v+\eta}{\boldsymbol{\epsilon}_{j,t}^v+\eta}, & \boldsymbol{\epsilon}_{j,t}^v < 0.5 \\ 0, & \boldsymbol{\epsilon}_{j,t}^v \geq 0.5 \end{cases}, \\ \tilde{\boldsymbol{\beta}}_{j,t+1}^v = \frac{\exp\left(\boldsymbol{\beta}_{j,t+1}^v\right)}{\sum_u \exp\left(\boldsymbol{\beta}_{j,t+1}^u\right)}, \end{aligned} \tag{10}$$

where $\boldsymbol{\beta}_{j,t+1}^v$ is the updated weight of the atom classifier that predicts the j-th label with $\boldsymbol{\mathcal{H}}^v$ at the t-th boosting iteration, and η is a small value to avoid numeric overflow. When the atom classifier is unable to give reliable prediction (*i.e.*, $\boldsymbol{\epsilon}_{j,t}^v \geq 0.5$), its weight will be reduced to construct the strong classifier, or it is easy to deteriorate the final prediction. After updating the weights of atom classifiers, the classification difficulty of samples *w.r.t.* each atom classifier

Algorithm 1 Training procedure of proposed DF^2-VB.

Input: MVML dataset $\mathcal{D} = \{(X_i, \boldsymbol{y}_i)\}_{i=1}^N$, the penalty coefficient α, number of epochs E.
Output: The multi-label classification model DF^2-VB.
Process:

```
for Training epoch e ≤ E do
    // Feature Extraction Phase
    for View number v ≤ V do
        Extract consensus representation C^v and complementarity representation P^v via Eq. (1);
    end for
    Decouple consensus and complementarity representations with Eq. (2).
    // Feature-level Fusion Phase
    for View number v ≤ V do
        Calculate membership degrees as feature-level weights S̃^v via Eq. (5).
    end for
    Obtain the comprehensive consensus representation Ĉ via Eq. (6).
    // Decision-level Fusion Phase
    for View number v ≤ V do
        Obtain the view representation H^v via Eq. (7);
    end for
    for Boosting iteration t ≤ T do
        Calculate error rates for all the atom classifiers via Eq. (9);
        Update weights for atom classifiers via Eq. (10);
        Update sample weights via Eq. (11).
    end for
    Calculate model predictions via Eq. (12);
    Update model parameters through the loss Eq. (14).
end for
```

Table 1. Performance comparisons between DF^2-VB and comparative models, where the best performance is shown in bold. For OE, RL, and Cov, lower values of them correspond to better performance. For AP and MiF1, higher values of them show better performance.

Datasets	Metrics	FIMAN	D-VSM	DIMC	ML-BVAE	VAMS	TMvML	DF^2-VB
Emotions	OE ↓	0.258±0.042	0.214±0.046	0.303±0.032	0.568±0.029	0.235±0.027	0.319±0.046	**0.206±0.024**
	RL ↓	0.161±0.026	0.137±0.015	0.183±0.019	0.423±0.035	0.144±0.013	0.184±0.019	**0.123±0.009**
	Cov ↓	7.796±0.189	1.624±0.038	1.873±0.037	3.163±0.242	1.659±0.055	1.914±0.066	**1.589±0.077**
	AP ↑	0.806±0.027	0.835±0.027	0.782±0.021	0.572±0.022	0.826±0.012	0.774±0.023	**0.840±0.012**
	MiF1 ↑	0.671±0.014	0.700±0.034	0.482±0.010	0.107±0.091	0.692±0.015	0.635±0.033	**0.724±0.025**
Yeast	OE ↓	0.257±0.013	**0.215±0.010**	0.248±0.009	0.253±0.012	0.222±0.008	0.250±0.010	0.220±0.018
	RL ↓	0.187±0.005	0.172±0.003	0.180±0.005	0.204±0.007	0.168±0.004	0.180±0.006	**0.158±0.006**
	Cov ↓	6.673±0.074	6.562±0.133	6.538±0.094	6.706±0.094	6.376±0.096	6.509±0.099	**6.154±0.092**
	AP ↑	0.740±0.007	0.760±0.005	0.738±0.006	0.712±0.007	0.763±0.009	0.745±0.007	**0.778±0.011**
	MiF1 ↑	0.111±0.008	0.655±0.008	0.468±0.003	0.479±0.008	0.656±0.012	0.577±0.007	**0.683±0.010**
Scene	OE ↓	0.280±0.018	0.189±0.016	0.263±0.009	0.464±0.032	0.198±0.014	0.207±0.011	**0.183±0.007**
	RL ↓	0.107±0.006	**0.058±0.007**	0.086±0.004	0.171±0.017	0.070±0.010	0.078±0.015	**0.058±0.002**
	Cov ↓	0.628±0.020	0.372±0.051	0.516±0.024	0.942±0.090	0.434±0.058	0.459±0.070	**0.370±0.013**
	AP ↑	0.827±0.010	0.890±0.011	0.844±0.004	0.717±0.023	0.878±0.014	0.795±0.004	**0.893±0.003**
	MiF1 ↑	0.616±0.008	0.777±0.015	0.308±0.002	0.304±0.095	0.748±0.016	0.713±0.023	**0.779±0.010**
Corel5k	OE ↓	0.489±0.017	0.410±0.018	0.613±0.019	0.626±0.019	0.432±0.016	0.575±0.014	**0.363±0.019**
	RL ↓	0.085±0.000	0.084±0.004	0.114±0.003	0.188±0.008	0.082±0.004	0.152±0.002	**0.068±0.004**
	Cov ↓	53.94±0.790	53.21±1.947	70.80±2.256	108.7±4.792	53.20±2.218	84.87±3.437	**42.65±2.184**
	AP ↑	0.430±0.007	0.475±0.008	0.333±0.008	0.286±0.000	0.452±0.009	0.369±0.006	**0.540±0.014**
	MiF1 ↑	0.361±0.009	0.399±0.004	0.029±0.000	0.025±0.001	0.372±0.005	0.414±0.008	**0.437±0.014**
Pascal	OE ↓	0.313±0.011	0.274±0.008	0.438±0.011	0.426±0.007	0.283±0.006	0.283±0.010	**0.269±0.012**
	RL ↓	0.118±0.003	0.077±0.001	**0.063±0.002**	0.106±0.001	0.101±0.002	0.111±0.006	0.066±0.004
	Cov ↓	3.486±0.081	2.342±0.044	**1.908±0.065**	2.966±0.028	2.231±0.038	2.482±0.022	2.094±0.096
	AP ↑	0.721±0.003	0.767±0.003	0.695±0.008	0.659±0.002	0.759±0.008	0.706±0.006	**0.784±0.009**
	MiF1 ↑	0.008±0.002	0.636±0.004	0.136±0.001	0.385±0.031	0.426±0.026	0.394±0.014	**0.648±0.011**
Espgame	OE ↓	0.628±0.004	**0.464±0.010**	0.615±0.013	0.615±0.010	0.470±0.011	0.690±0.023	**0.464±0.004**
	RL ↓	0.154±0.002	0.133±0.003	0.141±0.000	0.211±0.002	0.144±0.002	0.217±0.014	**0.123±0.001**
	Cov ↓	102.8±1.183	86.75±1.634	92.21±0.113	136.5±1.681	89.91±1.816	133.4±7.478	**82.39±0.751**
	AP ↑	0.284±0.002	0.364±0.003	0.273±0.001	0.257±0.001	0.266±0.002	0.223±0.012	**0.375±0.002**
	MiF1 ↑	0.242±0.002	0.332±0.003	0.034±0.000	0.008±0.002	0.320±0.000	0.022±0.001	**0.348±0.003**

becomes clear. We adjust the weights for samples according to the update rule that samples difficult to classify are assigned with higher weights while samples easy to classify are assigned with lower weights, which comes down to the following equation:

$$\boldsymbol{\omega}_{i,t+1} = \frac{\boldsymbol{\omega}_{i,t} \cdot \exp\left(-\boldsymbol{\beta}^v_{j,t+1} \boldsymbol{y}_{i,j} \hat{\boldsymbol{y}}^v_{i,j}\right)}{\sum_i \boldsymbol{\omega}_{i,t}}. \tag{11}$$

Finally, the probabilistic prediction of the j-th label $\bar{\boldsymbol{y}}_{i,j}$ aggregating the logits from all the views is given by

$$\bar{\boldsymbol{y}}_{i,j} = \sigma\left(\sum_v \tilde{\boldsymbol{\beta}}^v_{j,T} \cdot \tilde{\boldsymbol{o}}^v_{i,j}\right), \tag{12}$$

where T is the number of total boosting iterations. To train the joint classifier, we apply the binary cross-entropy as multi-label classification loss:

$$\mathcal{L}_{cls} = -\sum_{i=1}^{N}\sum_{j=1}^{L}\left[\boldsymbol{y}_{i,j}\log\bar{\boldsymbol{y}}_{i,j} + (1-\boldsymbol{y}_{i,j})\log(1-\bar{\boldsymbol{y}}_{i,j})\right]. \tag{13}$$

The overall loss function is given by

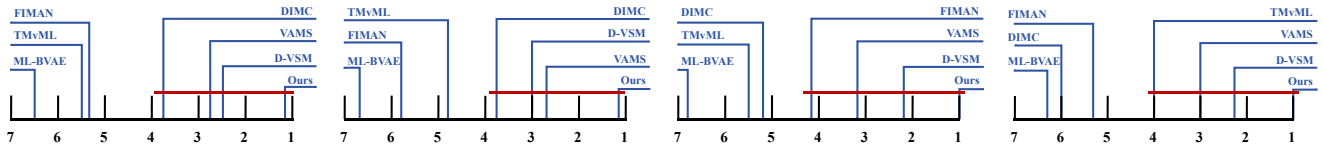
$$\mathcal{L} = \mathcal{L}_{cls} + \alpha\mathcal{L}_{de}, \tag{14}$$

where α is the penalty coefficient. During inference time, the weight for each atom classifier is determined, and Eq. (12) is employed to get final prediction. The details of the training algorithm of DF^2-VB is listed in Algorithm 1.

4. Experiment

4.1. Experimental Setting

We employ 6 public multi-view multi-label datasets to evaluate the model performance, including *Emotions*, *Scene*, *Yeast*, *Corel5k*, *Pascal*, and *Espgame*. For fair comparisons, we adopt 6 state-of-the-art the baseline models, including **FIMAN** [27], **D-VSM** [14], **DIMC** [26], **ML-BVAE** [3], **VAMS** [24], **TMvML** [39], and introduce widely-used Average Precision (AP), One Error (OE), Ranking Loss (RL), Coverage (Cov), and Micro F1 (MiF1) as metrics. Detailed definitions of these metrics are given in [21].



(a) Ranking Loss (b) Coverage (c) Average Precision (d) Micro F1

Figure 2. Experimental comparisons between the proposed model and all other comparative algorithms with the Bonferroni-Dunn test (CD = 3.290 at 0.05 significance level). Algorithms not connected with DF^2-VB are significantly inferior to DF^2-VB.

Table 2. Friedman statistics τ_F in terms of each evaluation metric (at 0.05 significance level).

Metrics	τ_F	Critical Difference Value
Ranking Loss	19.2056	
Coverage	13.9304	4.7233
Average Precision	38.4038	Methods: 7, Datasets: 6
Micro-F1	29.2012	

4.2. Multi-label Classification

In this section, we conduct the multi-label classification task to compare the performance gap between our proposed DF^2-VB and other advanced baselines. To obtain reliable results, five-fold cross-validation is performed on each dataset, and the means and standards of metrics are reported in Tab. 1. According to 210 experiment results, we have following observations:

(1) Our proposed DF^2-VB is superior to other baseline models in most cases across all metrics. Specifically, DF^2-VB is superior to ML-BVAE, FIMAN, TMvML, and VAMS in all cases, and also outperforms DIMC and D-VSM in 93.3% and 90.0% cases, respectively. These comparison results are strong proof to demonstrate that DF^2-VB is a promising MVMLC model.

(2) When compared with feature-level fusion methods, such as FIMAN and DIMC, DF^2-VB shows advantages in most metrics across all the datasets. These methods only focus on feature-level fusion, and they are limited in further enhancing the model performance due to the failure of fully leveraging the merits of supervision. However, DF^2-VB leverages supervision to strengthen the model prediction, attaining much better performance, especially MiF1.

(3) In comparison with decision-level fusion methods, such as ML-BVAE and TMvML, DF^2-VB shows significant improvements with all the evaluation metrics. These methods are not aware of the necessity to construct expressive representations from all the views. Instead, DF^2-VB fuses the view features through adaptive feature-level weights, which score the significance of feature elements to learn more expressive representations for accurate prediction.

In addition, we conduct a widely-used statistical test, *i.e.* Friedman test [2], to make comparisons for analyzing the relative performance among all comparative methods. As shown in Tab. 2, the null hypothesis of distinguishable performance among all comparative algorithms is rejected at 0.05 significance level. Furthermore, we employ the post-hoc Bonferroni-Dunn test [2] to compare the relative performance among the comparative algorithms. Fig. 2 provides the critical difference (CD) diagrams on evaluation metrics, where the average rank of each algorithm is marked along the axis. According to Fig. 2, it is observed that DF^2-VB has the best performance across all metrics.

In summary, the proposed DF^2-VB combines the feature-level and decision-level fusion for expressive and discriminative representations, which consistently displays the best performance against existing MVMLC methods.

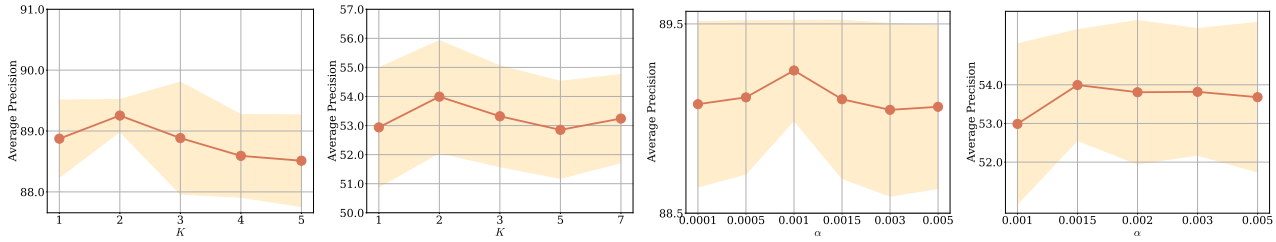
4.3. Ablation Study

In the ablation study, we evaluate the effectiveness of each component by comparing DF^2-VB with four degenerated variants: (1) DF^2-VB without FDF and VB, *i.e.*, a model directly uses the averaged consensus representations from all the views for prediction; (2) DF^2-VB without VB, *i.e.*, a model uses the comprehensive consensus representation $\hat{C}$ with FDF for prediction; (3) DF^2-VB without FDF, *i.e.*, a model uses the averaged consensus representations with VB for prediction; (4) DF^2-VB without DL, *i.e.*, a model without the decoupling loss.

The comparison results between DF^2-VB and other variants are provided in Tab. 3. From the results, we find that each component contributes to the model performance improvement, and this shows the functions of all the components. Additionally, the performance of the model without VB is inferior to that without FDF in general, and this implies that the feature-level fusion is more critical than the decision-level fusion. Finally, we find that the proposed DF^2-VB has the best performance among other variants, which illustrates that both feature-level fusion and decision-level fusion are mutually beneficial and complementary. This complementarity between both fusions is strong proof of our motivation.

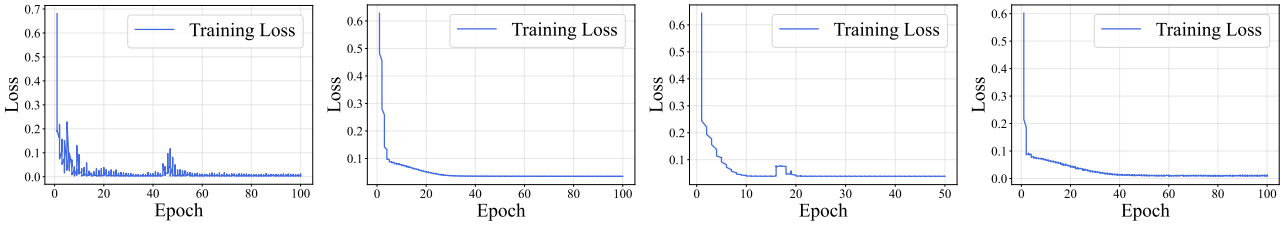
4.4. Sensitivity Analysis

In sensitivity analysis, we explore the relationship between the model performance and two key hyperparameters, namely the number of Gaussian membership functions



(a) Scene (b) Corel5K (c) Scene (d) Corel5K

Figure 3. Hyperparameter sensitivity analysis about K and α.



(a) Scene (b) Corel5K (c) Pascal (d) Espgame

Figure 4. Convergence of loss functions for different datasets.

Table 3. Ablation study between DF^2-VB and four variants.

Emotions	Average Precision	One Error	Ranking Loss	Coverage	Micro-F1
W/o FDF & VB	0.828 ± 0.017	0.238 ± 0.023	0.135 ± 0.016	1.644 ± 0.096	0.595 ± 0.012
W/o VB	0.829 ± 0.011	0.231 ± 0.021	0.133 ± 0.009	1.633 ± 0.057	0.633 ± 0.011
W/o FDF	0.834 ± 0.014	0.221 ± 0.028	0.131 ± 0.013	1.626 ± 0.090	0.716 ± 0.014
W/o DL	0.838 ± 0.012	0.209 ± 0.022	**0.123 ± 0.012**	1.599 ± 0.072	0.722 ± 0.015
DF^2-VB	**0.840 ± 0.012**	**0.206 ± 0.024**	**0.123 ± 0.009**	**1.589 ± 0.077**	**0.724 ± 0.025**
Scene	Average Precision	One Error	Ranking Loss	Coverage	Micro-F1
W/o FDW & VB	0.884 ± 0.008	0.194 ± 0.012	0.069 ± 0.008	0.430 ± 0.046	0.767 ± 0.011
W/o VB	0.888 ± 0.006	0.188 ± 0.011	0.064 ± 0.004	0.406 ± 0.027	0.774 ± 0.009
W/o FDF	0.890 ± 0.004	0.190 ± 0.007	0.060 ± 0.004	0.384 ± 0.027	0.775 ± 0.011
W/o DL	0.892 ± 0.009	0.186 ± 0.015	0.059 ± 0.006	0.374 ± 0.031	0.776 ± 0.010
DF^2-VB	**0.893 ± 0.003**	**0.183 ± 0.007**	**0.058 ± 0.002**	**0.370 ± 0.013**	**0.779 ± 0.010**

K and the penalty coefficient α. We provide the varying hyperparameter configurations and the corresponding model performance in Fig. 3, where we observe that both hyperparameters affect the model performance. When the hyperparameters increase, the model performance gradually improves at first and then drops later. In our experiments, the optimal values of K and α for Scene and Corel5K datasets are $\{2, 2\}$ and $\{0.001, 0.0015\}$, respectively.

4.5. Convergence Analysis

We conduct the convergence analysis of the proposed method across four datasets, including Scene, Corel5K, Pascal, and Espgame. And the changes of loss values during training are provided in Fig. 4. We observe that the loss values decrease at first and then become steady across datasets. This phenomenon empirically illustrates that DF^2-VB is convergent during the training process.

5. Conclusion

In this paper, we propose DF^2-VB, a unified dual-level fusion MVMLC model that jointly exploits complementary strengths of feature-level and decision-level fusion to mitigate their respective weaknesses, namely the limited discriminability and the insufficient representation expressiveness. By jointly incorporating FDF to enhance feature-level compatibility and VB to reinforce decision-level collaboration, DF^2-VB effectively leverages both consensus and complementary information across heterogeneous view data to enhance model performance. Extensive experiments on multiple datasets demonstrate that DF^2-VB consistently outperforms advanced methods, confirming that the design of dual-level fusion is successful.

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